Linear Algebra: Vectors, Matrices, and Transformations

A Geometric and Algebraic Perspective

Your Name

Your Institution

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Presentation Outline

Vectors: Addition and Scalar Multiplication

Definition (Vectors and Coordinates)

A **vector** is a geometric object with magnitude and direction, represented in a coordinate system as an array of numbers (components).

Derivation / Rules

Vector addition is performed element-wise. Scalar multiplication distributes the scalar to each individual component:

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \pm \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 \pm x_2 \\ y_1 \pm y_2 \end{bmatrix} \quad \text{and} \quad c \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} cx \\ cy \end{bmatrix}$$

Vectors: Addition and Scalar Multiplication (Example)

Example Problem

Problem: Let $\vec{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Find $2\vec{u} + \vec{v}$.

Solution:

$$\begin{aligned} 2\vec{u} + \vec{v} &= 2\begin{bmatrix} 2 \\ -1 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 4+3 \\ -2+4 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix} \end{aligned}$$

Linear Combinations, Span, and Basis Vectors

Definition (Linear Combination and Basis)

A **linear combination** is the sum of scaled vectors: $c_1\vec{v}_1 + \cdots + c_n\vec{v}_n$. The **span** is the set of all possible linear combinations. **Basis vectors** are a linearly independent set that spans the space.

Derivation / Concept

The standard 2D basis vectors are $\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Any vector $\vec{w}$ in $\mathbb{R}^2$ can be uniquely derived as $x\hat{i} + y\hat{j}$.

Linear Combinations, Span, and Basis Vectors (Example)

Example Problem

Problem: Express the vector $\vec{w} = \begin{bmatrix} 5 \\ -7 \end{bmatrix}$ as a linear combination of the standard basis vectors $\hat{i}$ and $\hat{j}$.

Solution:

$$\begin{aligned} \vec{w} &= \begin{bmatrix} 5 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -7 \end{bmatrix} \\ &= 5 \begin{bmatrix} 1 \\ 0 \end{bmatrix} - 7 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= 5\hat{i} - 7\hat{j} \end{aligned}$$

Matrix Multiplication as Composition

Definition (Linear Transformation and Composition)

A **linear transformation** is a mapping that preserves vector addition and scalar multiplication (grid lines remain parallel/evenly spaced, origin is fixed). **Matrix composition** applies successive transformations via matrix multiplication.

Derivation

Applying a transformation matrix A to a vector $\vec{x}$ is computed by taking the dot product of A's rows with $\vec{x}$:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

Matrix Multiplication as Composition (Example)

Example Problem

Problem: Apply the shear transformation matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ to the vector $\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Solution:

$$\begin{aligned} A\vec{v} &= \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(2) \\ 0(1) + 1(2) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 4 \\ 0 + 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix} \end{aligned}$$

Three-Dimensional and Non-square Matrices

Definition (Dimension Transformations)

Non-square matrices represent linear transformations between spaces of differing dimensions. An $m \times n$ matrix transforms an n-dimensional space into an m-dimensional space.

Derivation

Mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ uses an $m \times n$ matrix A. The product $A\vec{x}$ computes the coordinates in the target m-dimensional space, essentially taking a linear combination of the n column vectors of A (which live in $\mathbb{R}^m$).

Three-Dimensional and Non-square Matrices (Example)

Example Problem

Problem: Map a 2D vector $\vec{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ into 3D space using the transformation matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{bmatrix}$.

Solution:

$$
\begin{aligned}
A\vec{x} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \\
&= 2 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \\
&= \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}
\end{aligned}
$$

The Determinant and Inverse Matrices

Definition (Determinant and Inverse)

The **determinant** (det) is the geometric scaling factor of a transformation's area or volume. The **inverse matrix** (A^{-1}) perfectly undoes the transformation A, such that $A^{-1}A = I$. An inverse only exists if $\det(A) \neq 0$.

Derivation for 2x2 Matrices

For a matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

Determinant: $\det(A) = ad - bc$.

Inverse: $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

The Determinant and Inverse Matrices (Example)

Example Problem

Problem: Find the inverse of $A = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$.

Solution:

1. Calculate the determinant:

$$\det(A) = (4)(2) - (3)(3) = 8 - 9 = -1$$

2. Apply the inverse formula:

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 2 & -3 \\ -3 & 4 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -2 & 3 \\ 3 & -4 \end{bmatrix}$$

Column Space, Null Space, and Cramer's Rule

Definition (Spaces and Cramer's Rule)

The **column space** is the span of a matrix's columns. The **null space** contains all vectors mapped to the zero vector. **Cramer's Rule** is a theorem giving an explicit determinant-based solution for systems of linear equations $A\vec{x} = \vec{b}$.

Derivation of Cramer's Rule

To solve for the i-th variable x_i: substitute the i-th column of the coefficient matrix A with the target vector $\vec{b}$ to form a new matrix A_i. Then calculate:

$$x_i = \frac{\det(A_i)}{\det(A)}$$

Cramer's Rule (Example)

Example Problem

Problem: Solve the system $2x + y = 5$ and $x - y = 1$ using Cramer's rule.

Solution: $A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$. $\det(A) = -2 - 1 = -3$.

Find x:

$$A_x = \begin{bmatrix} 5 & 1 \\ 1 & -1 \end{bmatrix} \implies \det(A_x) = -5 - 1 = -6 \implies x = \frac{-6}{-3} = 2$$

Find y:

$$A_y = \begin{bmatrix} 2 & 5 \\ 1 & 1 \end{bmatrix} \implies \det(A_y) = 2 - 5 = -3 \implies y = \frac{-3}{-3} = 1$$

Dot Products and Duality

Definition (Dot Product and Duality)

The **dot product** measures directional alignment between vectors. **Duality** highlights that any linear transformation mapping 2D vectors to a 1D number line exactly matches taking a dot product with a specific 2D vector.

Derivation

Algebraically, it is the sum of the products of corresponding entries:

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Geometrically, it relates to the angle θ between vectors:

$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos(\theta)$$

Dot Products and Duality (Example)

Example Problem

Problem: Calculate the dot product of $\vec{u} = \begin{bmatrix} 3 \\ -2 \\ 4 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$.

Solution:

$$\begin{aligned} \vec{u} \cdot \vec{v} &= (3)(1) + (-2)(5) + (4)(2) \\ &= 3 - 10 + 8 \\ &= 1 \end{aligned}$$

Note: Since the dot product is positive, the angle between the vectors is acute.

Cross Products and Linear Transformations

Definition (Cross Product)

The **cross product** $\vec{u} \times \vec{v}$ produces a vector that is orthogonal (perpendicular) to both $\vec{u}$ and $\vec{v}$ in 3D space. Its magnitude is exactly equal to the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$.

Derivation via Determinant

The cross product is formally calculated using a pseudo-determinant of the basis vectors:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$$

Cross Products and Linear Transformations (Example)

Example Problem

Problem: Find $\vec{u} \times \vec{v}$ for $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ (which is $\hat{i}$) and $\vec{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ (which is $\hat{j}$).

Solution:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix}$$

$$= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(1-0)$$

$$= \hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Vector Spaces and Change of Basis

Definition (Vector Spaces and Change of Basis)

A **vector space** is any set of objects obeying 8 strict axioms for vector addition and scalar multiplication. **Change of basis** translates vectors and transformations from our standard coordinate system into someone else's basis vectors.

Derivation

Let P be a matrix whose columns are the new basis vectors.

- To express a standard vector $\vec{v}$ in the new basis P, compute $[\vec{v}]_P = P^{-1}\vec{v}$.
- To translate a transformation matrix A into the new basis, use $A' = P^{-1}AP$.

Vector Spaces and Change of Basis (Example)

Example Problem

Problem: Express the vector $\vec{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ (in standard basis) in terms of a new basis defined by the columns of $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

Solution: 1. Find P^{-1}:

$$P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

2. Compute $[\vec{v}]_P = P^{-1}\vec{v}$:

$$[\vec{v}]_P = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 - 2 \\ 0 + 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Eigenvectors and Eigenvalues

Definition (Eigenvectors and Eigenvalues)

Eigenvectors ($\vec{v}$) are special, non-zero vectors that remain on their original span during a linear transformation A (they only scale, they do not rotate). The scalar by which they stretch or compress is the **eigenvalue** (λ).

Derivation (Characteristic Equation)

The relationship is defined as $A\vec{v} = \lambda\vec{v}$, which rewrites to $(A - \lambda I)\vec{v} = \vec{0}$. To find non-trivial solutions for $\vec{v}$, we solve the characteristic equation:

$$\det(A - \lambda I) = 0$$

Eigenvectors and Eigenvalues (Example - Part 1)

Example Problem

Problem: Find the eigenvalues and corresponding eigenvectors of the matrix $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.

Solution (Part 1 - Eigenvalues):

1. Set up the characteristic equation $\det(A - \lambda I) = 0$:

$$\det\left(\begin{bmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix}\right) = 0$$

2. Calculate the determinant:

$$(3-\lambda)(2-\lambda) - (1)(0) = 0$$
$$(3-\lambda)(2-\lambda) = 0$$

3. Solve for λ: The eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 2$.

Eigenvectors and Eigenvalues (Example - Part 2)

Solution Continued (Part 2 - Eigenvectors)

For $\lambda_1 = 3$**:** Solve $(A - 3I)\vec{v} = \vec{0}$

$$\begin{bmatrix} 3-3 & 1 \\ 0 & 2-3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This yields $y = 0$, with x as a free variable. Setting $x = 1$, the eigenvector is $\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

For $\lambda_2 = 2$**:** Solve $(A - 2I)\vec{v} = \vec{0}$

$$\begin{bmatrix} 3-2 & 1 \\ 0 & 2-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This yields $x + y = 0 \implies x = -y$. Setting $y = 1$, the eigenvector is $\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

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